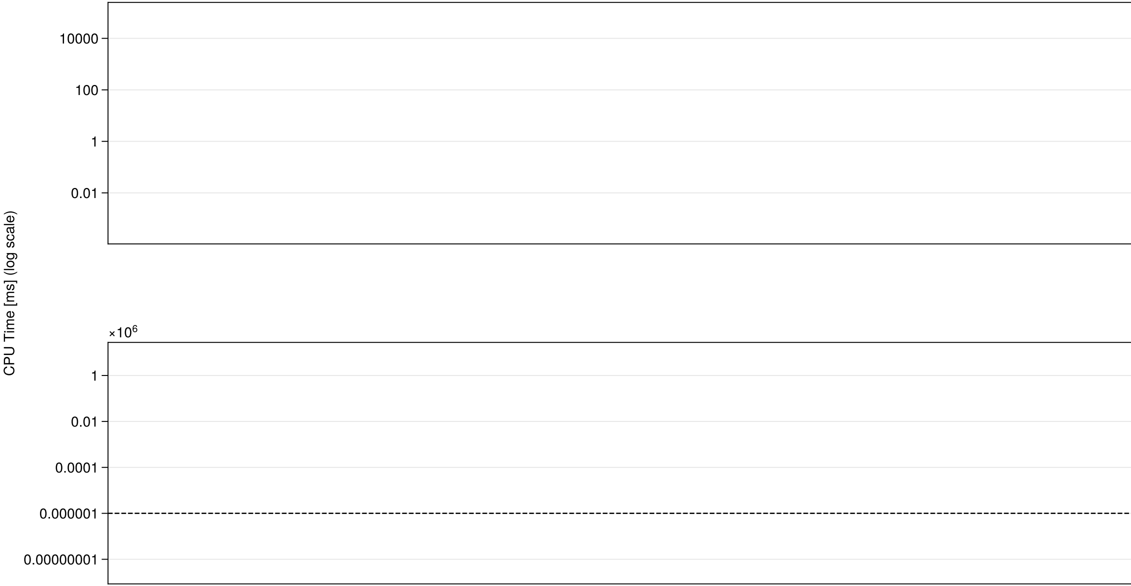
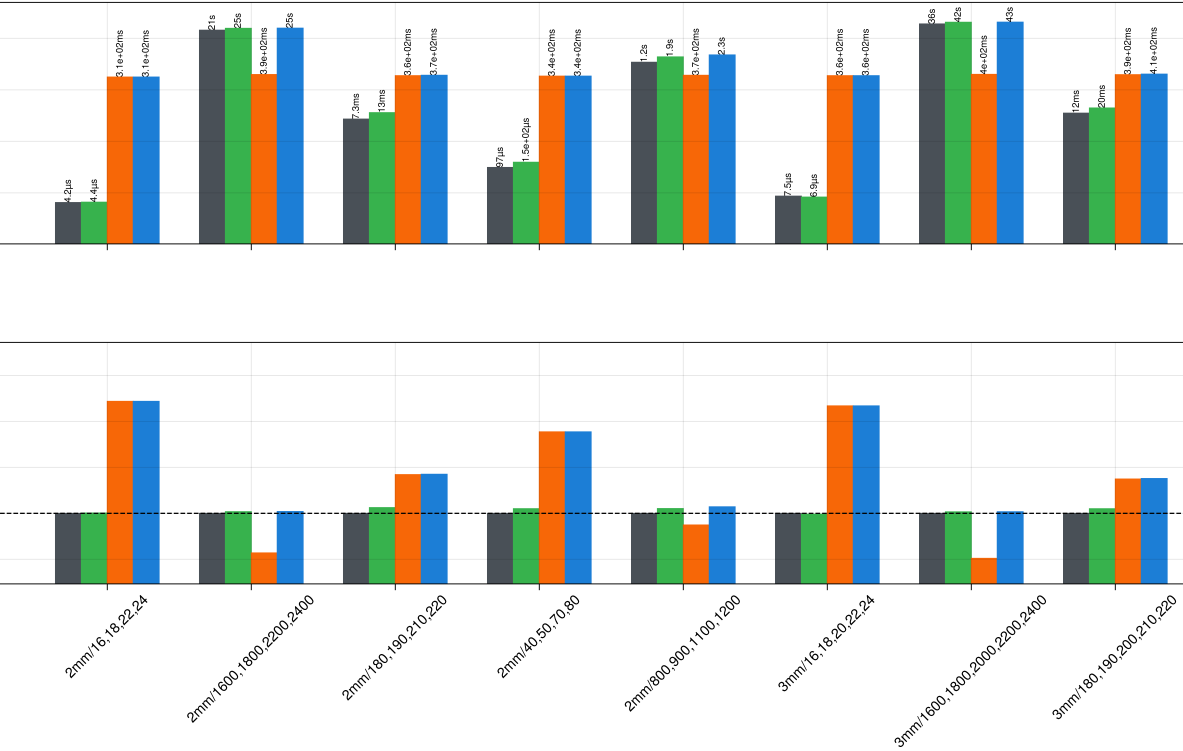
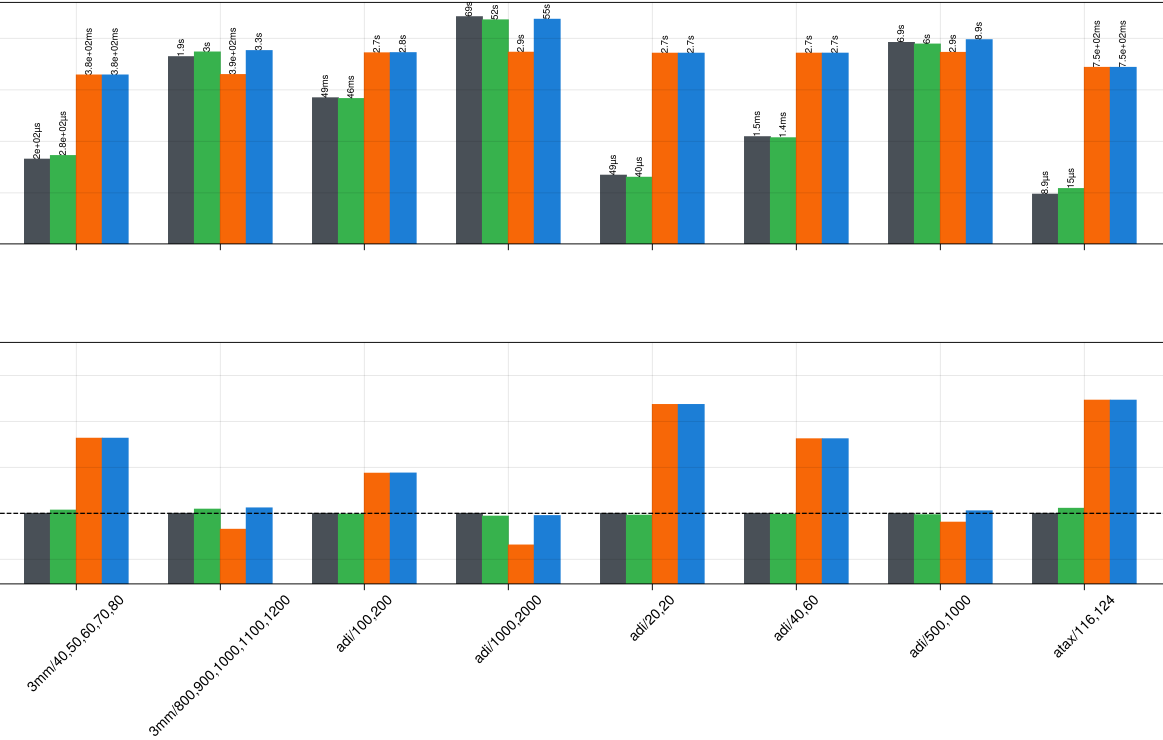
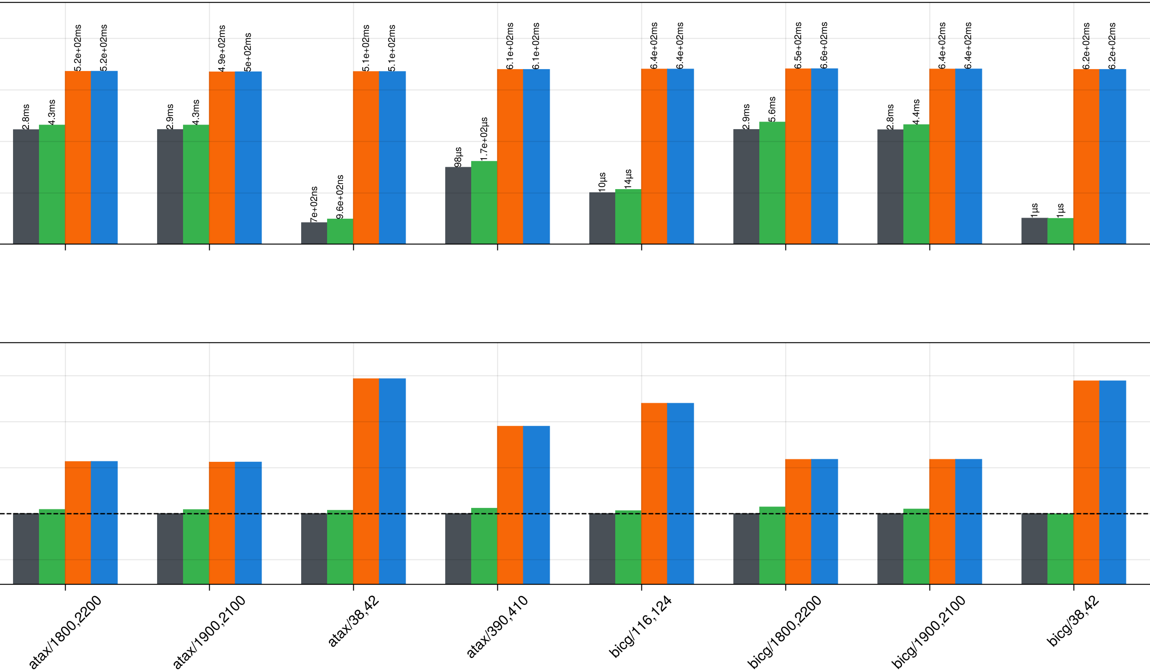
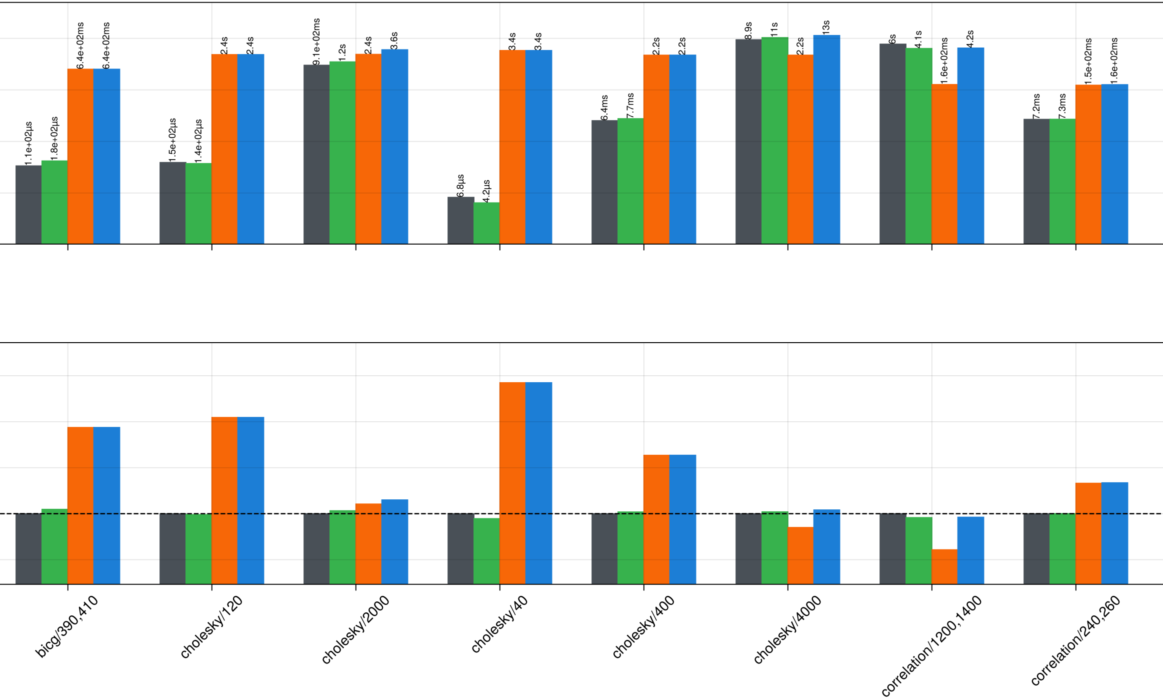
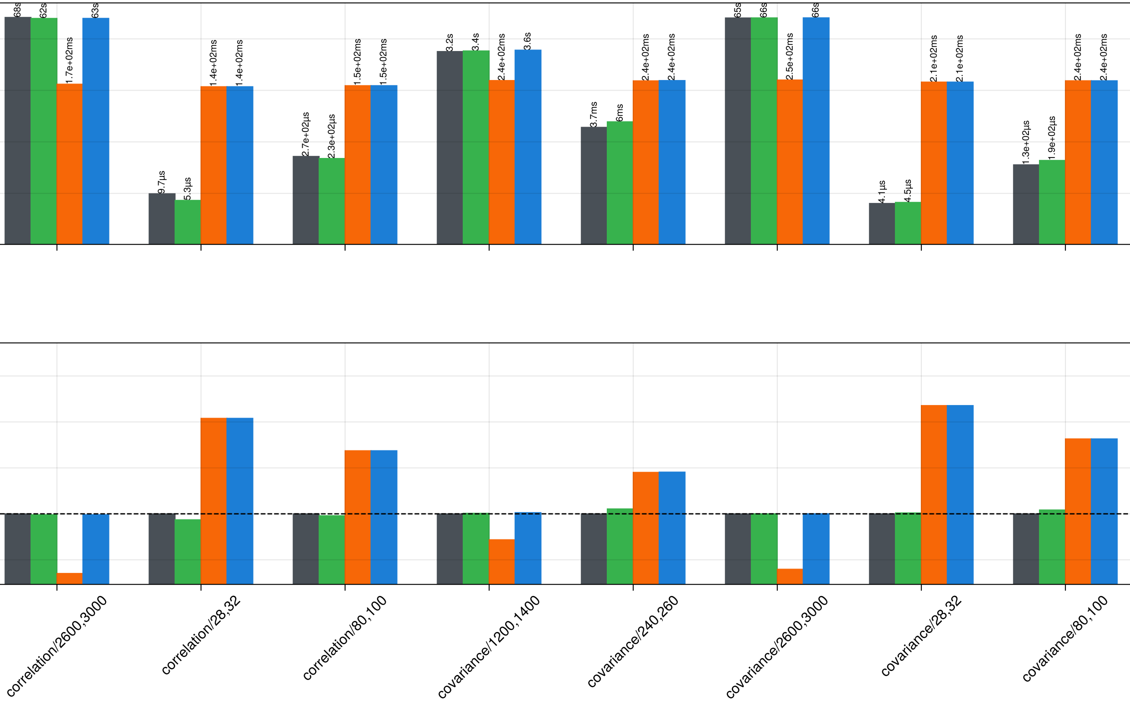

CPU Time [ms] (log scale)
10000
100
1
0.01
×10^6
1
0.01
0.0001
0.000001
0.00000001

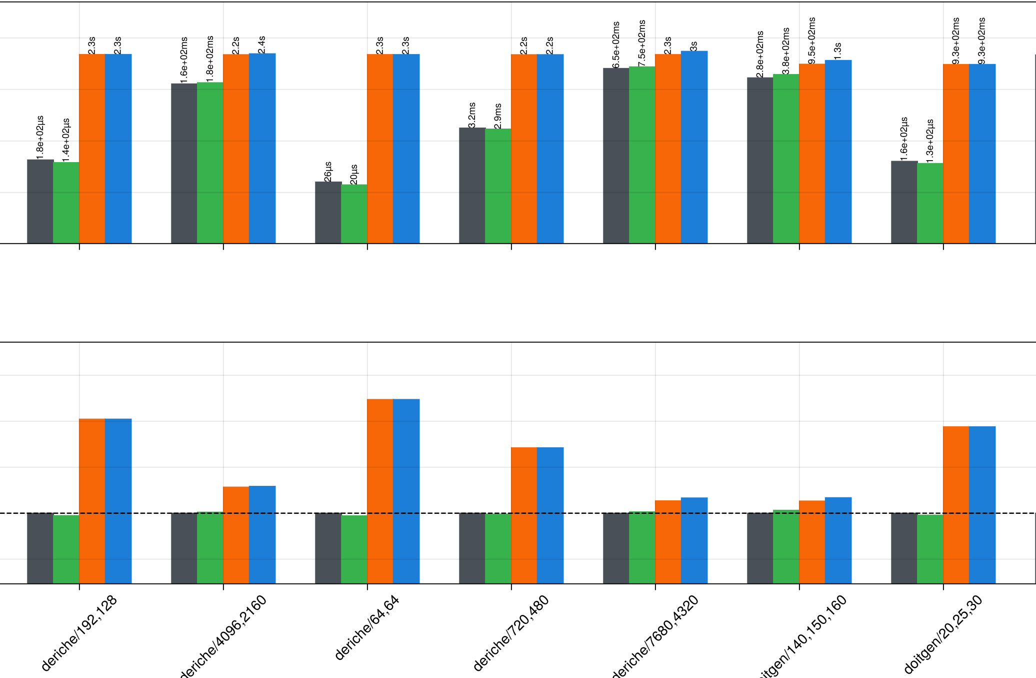

4.2µs
4.4µs
3.1e+02ms
3.1e+02ms
21s
25s
3.9e+02ms
25s
7.3ms
13ms
3.6e+02ms
3.7e+02ms
97µs
1.5e+02µs
3.4e+02ms
3.4e+02ms
1.2s
1.9s
3.7e+02ms
2.3s
7.5µs
6.9µs
3.6e+02ms
3.6e+02ms
36s
42s
4e+02ms
43s
12ms
20ms
3.9e+02ms
4.1e+02ms
2mm/16,18,22,24
2mm/1600,1800,2200,2400
2mm/180,190,210,220
2mm/40,50,70,80
2mm/800,900,1100,1200
3mm/16,18,20,22,24
3mm/1600,1800,2000,2200,2400
3mm/180,190,200,210,220

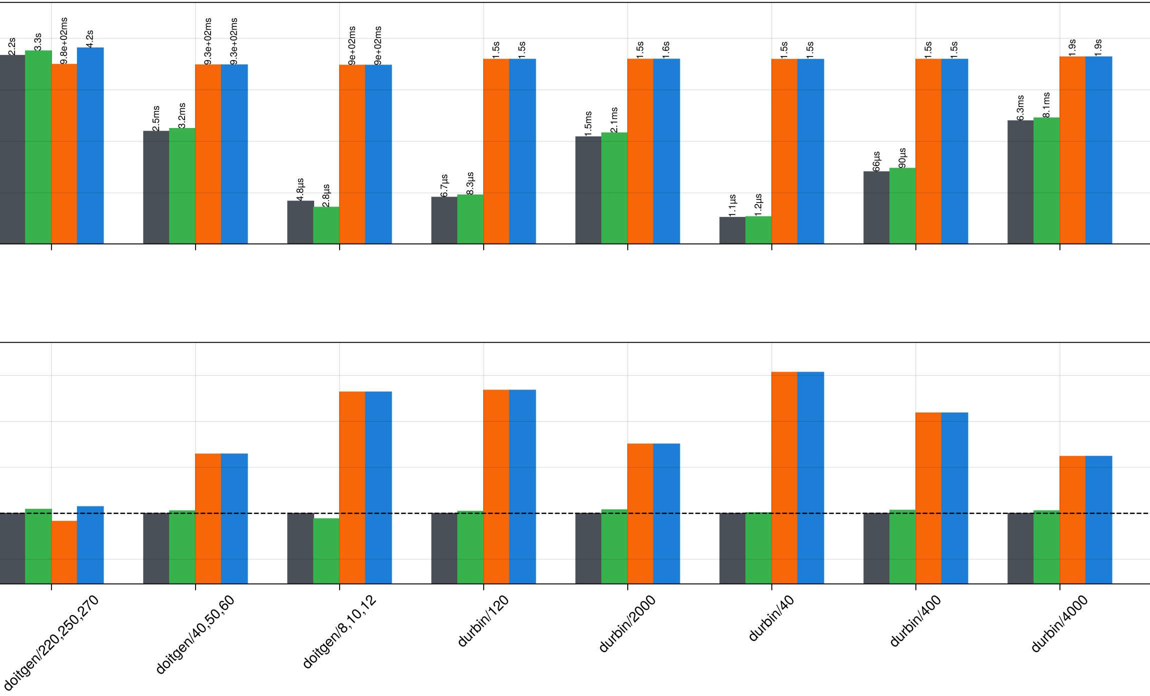

2e+02µs
2.8e+02µs
3.8e+02ms
3.8e+02ms
1.9s
3s
3.9e+02ms
3.3s
49ms
46ms
2.7s
2.8s
69s
52s
2.9s
55s
49µs
40µs
2.7s
2.7s
1.5ms
1.4ms
2.7s
2.7s
6.9s
6s
2.9s
8.9s
8.9µs
15µs
7.5e+02ms
7.5e+02ms
3mm/40,50,60,70,80
3mm/800,900,1000,1100,1200
adi/100,200
adi/1000,2000
adi/20,20
adi/40,60
adi/500,1000
atax/116,124

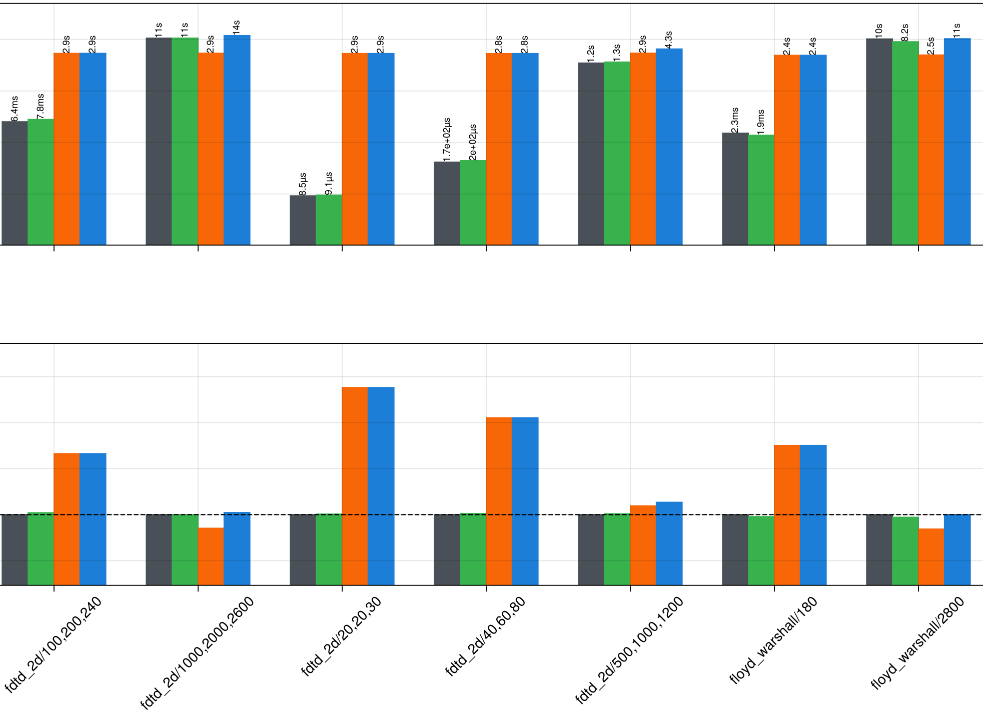

2.8ms
4.3ms
5.2e+02ms
5.2e+02ms
2.9ms
4.3ms
4.9e+02ms
5e+02ms
7e+02ns
9.6e+02ns
5.1e+02ms
5.1e+02ms
98µs
1.7e+02µs
6.1e+02ms
6.1e+02ms
10µs
14µs
6.4e+02ms
6.4e+02ms
2.9ms
5.6ms
6.5e+02ms
6.6e+02ms
2.8ms
4.4ms
6.4e+02ms
6.4e+02ms
1µs
1µs
6.2e+02ms
6.2e+02ms
atax/1800,2200
atax/1900,2100
atax/38,42
atax/390,410
bicg/116,124
bicg/1800,2200
bicg/1900,2100
bicg/38,42

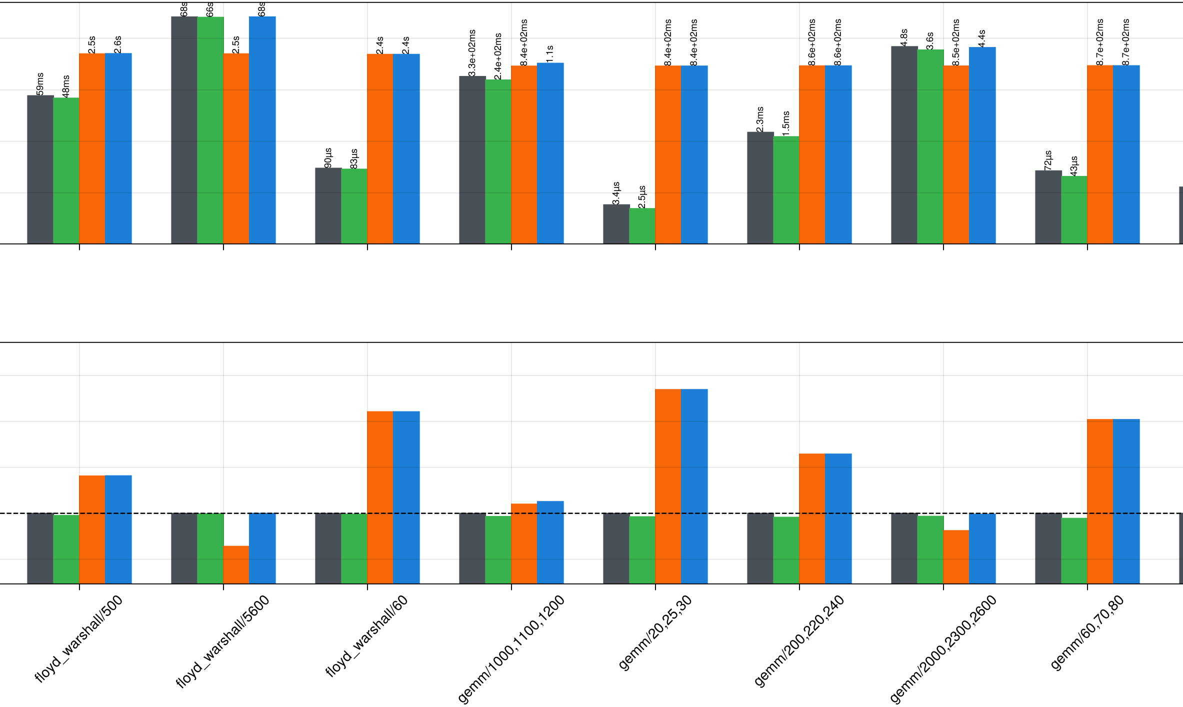

1.1e+02µs
1.8e+02µs
6.4e+02ms
6.4e+02ms
1.5e+02µs
1.4e+02µs
2.4s
2.4s
9.1e+02ms
1.2s
2.4s
3.6s
6.8µs
4.2µs
3.4s
3.4s
6.4ms
7.7ms
2.2s
2.2s
8.9s
11s
2.2s
13s
6s
4.1s
1.6e+02ms
4.2s
7.2ms
7.3ms
1.5e+02ms
1.6e+02ms
bicg/390,410
cholesky/120
cholesky/2000
cholesky/40
cholesky/400
cholesky/4000
correlation/1200,1400
correlation/240,260

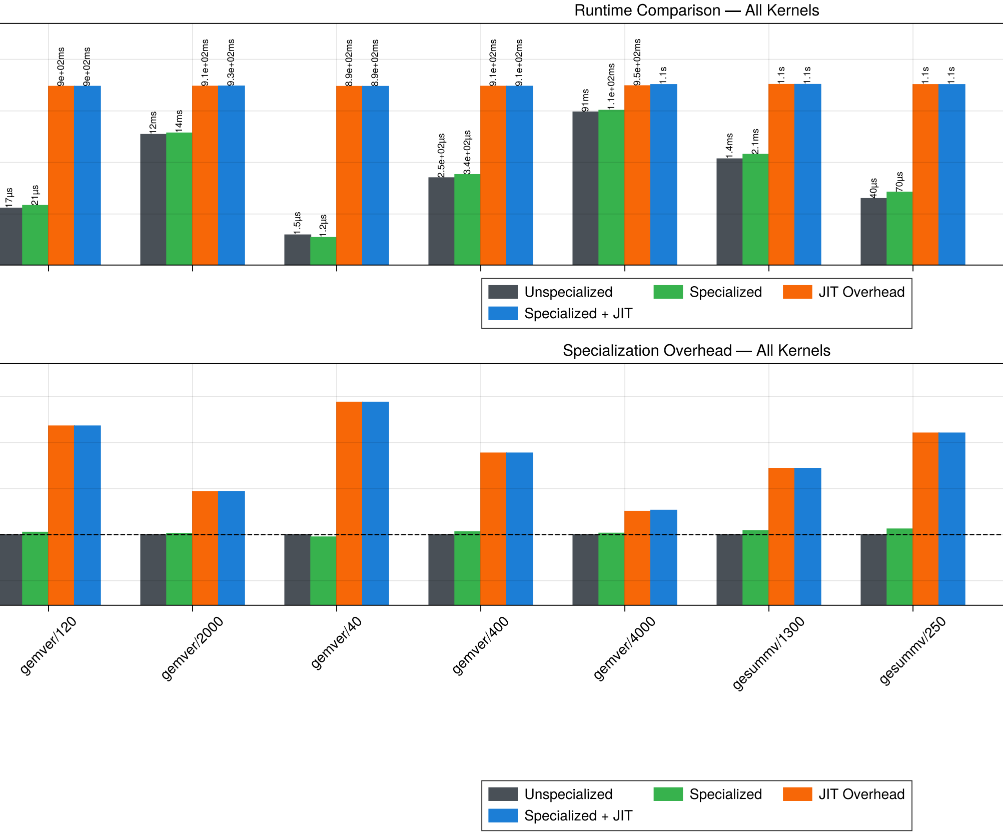

68s
62s
1.7e+02ms
63s
9.7µs
5.3µs
1.4e+02ms
1.4e+02ms
2.7e+02µs
2.3e+02µs
1.5e+02ms
1.5e+02ms
3.2s
3.4s
2.4e+02ms
3.6s
3.7ms
6ms
2.4e+02ms
2.4e+02ms
65s
66s
2.5e+02ms
66s
4.1µs
4.5µs
2.1e+02ms
2.1e+02ms
1.3e+02µs
1.9e+02µs
2.4e+02ms
2.4e+02ms
correlation/2600,3000
correlation/28,32
correlation/80,100
covariance/1200,1400
covariance/240,260
covariance/2600,3000
covariance/28,32
covariance/80,100

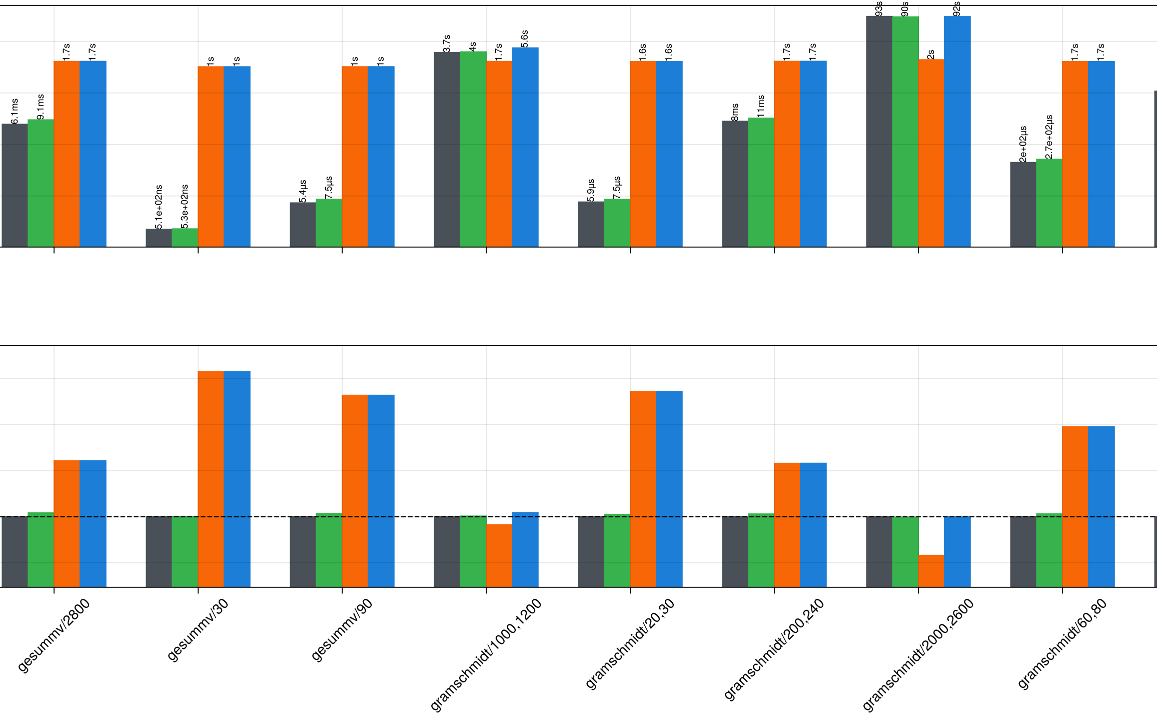

1.8e+02µs
1.4e+02µs
2.3s
2.3s
1.6e+02ms
1.8e+02ms
2.2s
2.4s
26µs
20µs
2.3s
2.3s
3.2ms
2.9ms
2.2s
2.2s
6.5e+02ms
7.5e+02ms
2.3s
3s
2.8e+02ms
3.8e+02ms
9.5e+02ms
1.3s
1.6e+02µs
1.3e+02µs
9.3e+02ms
9.3e+02ms
deriche/192,128
deriche/4096,2160
deriche/64,64
deriche/720,480
deriche/7680,4320
doitgen/140,150,160
doitgen/20,25,30

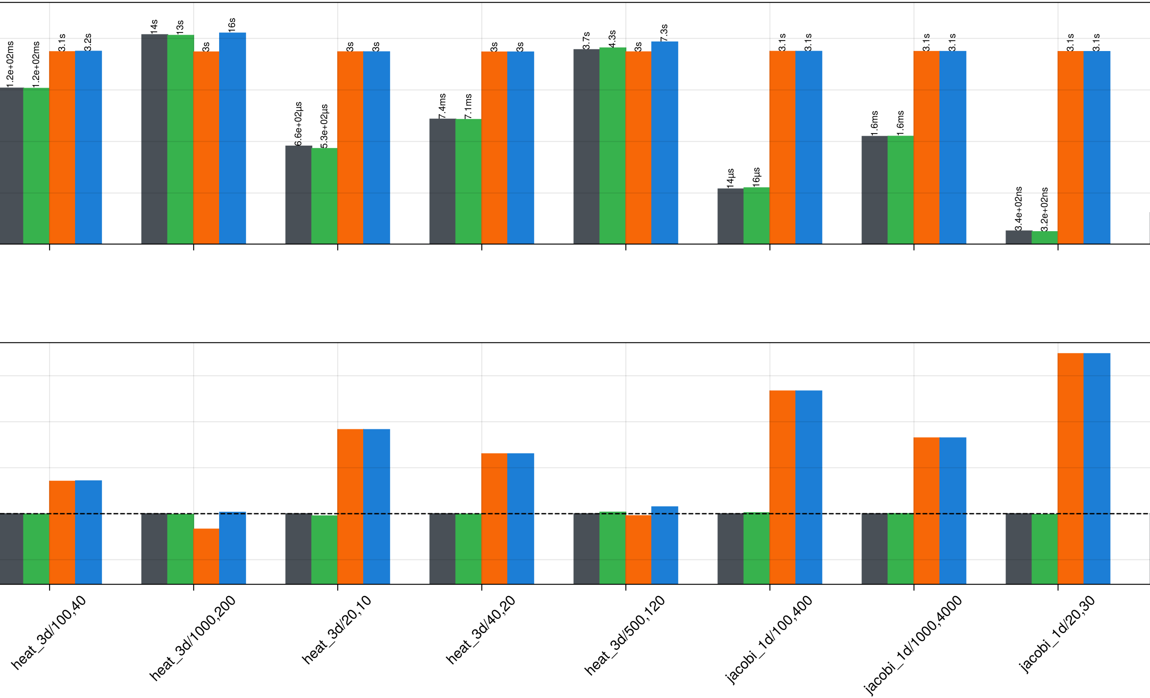

2.2s
3.3s
9.8e+02ms
4.2s
2.5ms
3.2ms
9.3e+02ms
9.3e+02ms
4.8µs
2.8µs
9e+02ms
9e+02ms
6.7µs
8.3µs
1.5s
1.5s
1.5ms
2.1ms
1.5s
1.6s
1.1µs
1.2µs
1.5s
1.5s
66µs
90µs
1.5s
1.5s
6.3ms
8.1ms
1.9s
1.9s
doitgen/220,250,270
doitgen/40,50,60
doitgen/8,10,12
durbin/120
durbin/2000
durbin/40
durbin/400
durbin/4000

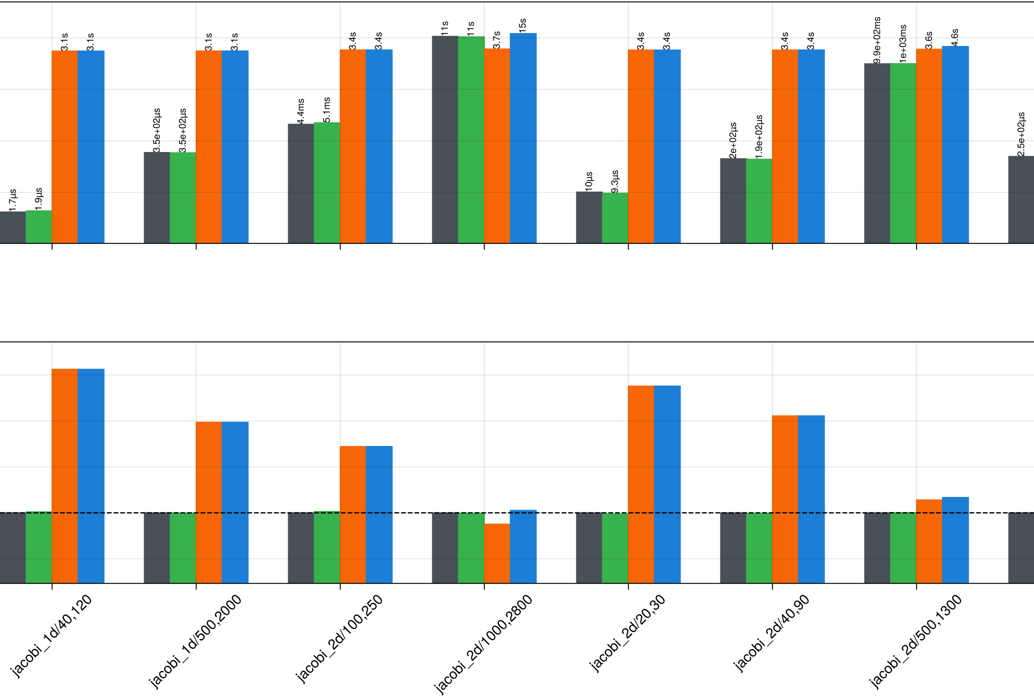

6.4ms
7.8ms
2.9s
2.9s
11s
11s
2.9s
14s
8.5µs
9.1µs
2.9s
2.9s
1.7e+02µs
2e+02µs
2.8s
2.8s
1.2s
1.3s
2.9s
4.3s
2.3ms
1.9ms
2.4s
2.4s
10s
8.2s
2.5s
11s
fdtd_2d/100,200,240
fdtd_2d/1000,2000,2600
fdtd_2d/20,20,30
fdtd_2d/40,60,80
fdtd_2d/500,1000,1200
floyd_warshall/180
floyd_warshall/2800

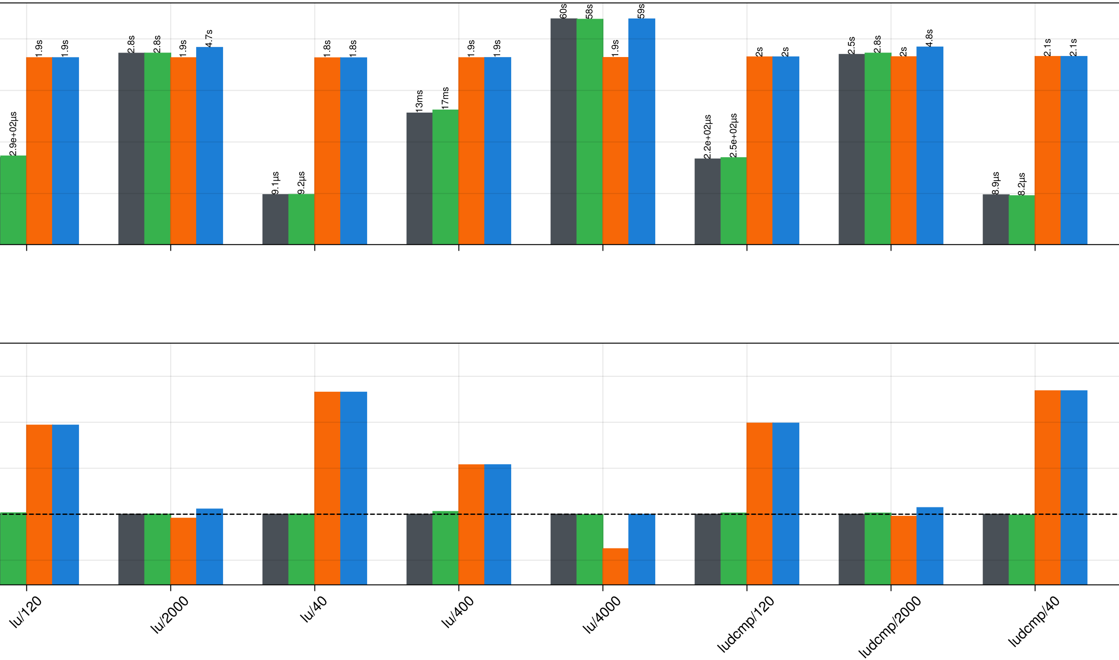

59ms
48ms
2.5s
2.6s
68s
66s
2.5s
68s
90µs
83µs
2.4s
2.4s
3.3e+02ms
2.4e+02ms
8.4e+02ms
1.1s
3.4µs
2.5µs
8.4e+02ms
8.4e+02ms
2.3ms
1.5ms
8.6e+02ms
8.6e+02ms
4.8s
3.6s
8.5e+02ms
4.4s
72µs
43µs
8.7e+02ms
8.7e+02ms
floyd_warshall/500
floyd_warshall/5600
floyd_warshall/60
gemm/1000,1100,1200
gemm/20,25,30
gemm/200,220,240
gemm/2000,2300,2600
gemm/60,70,80

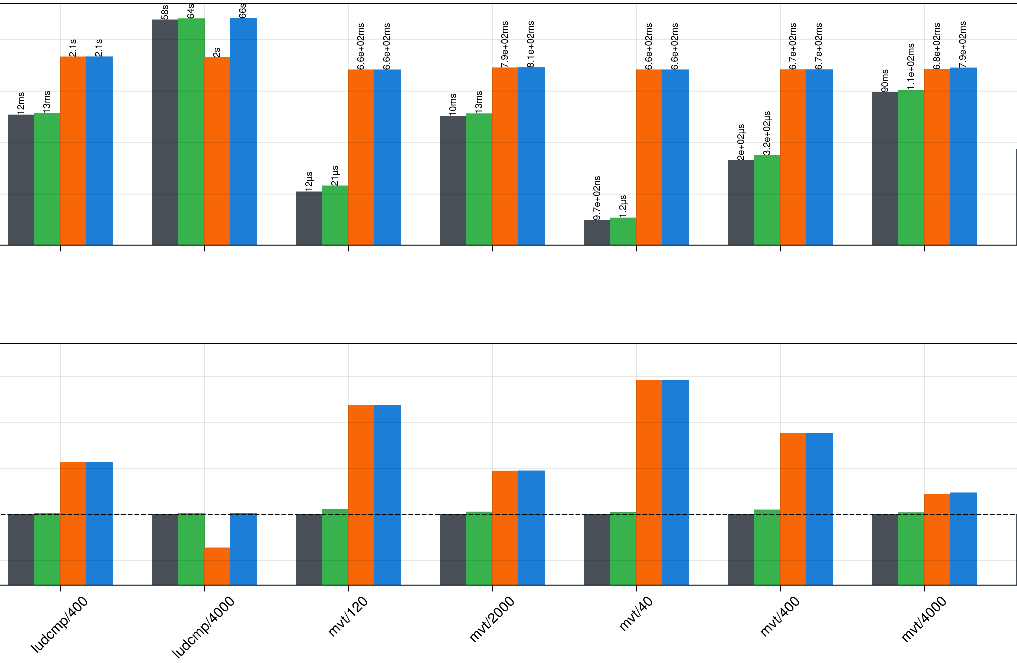

Runtime Comparison — All Kernels
17µs
21µs
9e+02ms
9e+02ms
12ms
14ms
9.1e+02ms
9.3e+02ms
1.5µs
1.2µs
8.9e+02ms
8.9e+02ms
2.5e+02µs
3.4e+02µs
9.1e+02ms
9.1e+02ms
91ms
1.1e+02ms
9.5e+02ms
1.1s
1.4ms
2.1ms
1.1s
1.1s
40µs
70µs
1.1s
1.1s
Unspecialized
Specialized
JIT Overhead
Specialized + JIT
Specialization Overhead — All Kernels
gemver/120
gemver/2000
gemver/40
gemver/400
gemver/4000
gesummv/1300
gesummv/250
Unspecialized
Specialized
JIT Overhead
Specialized + JIT

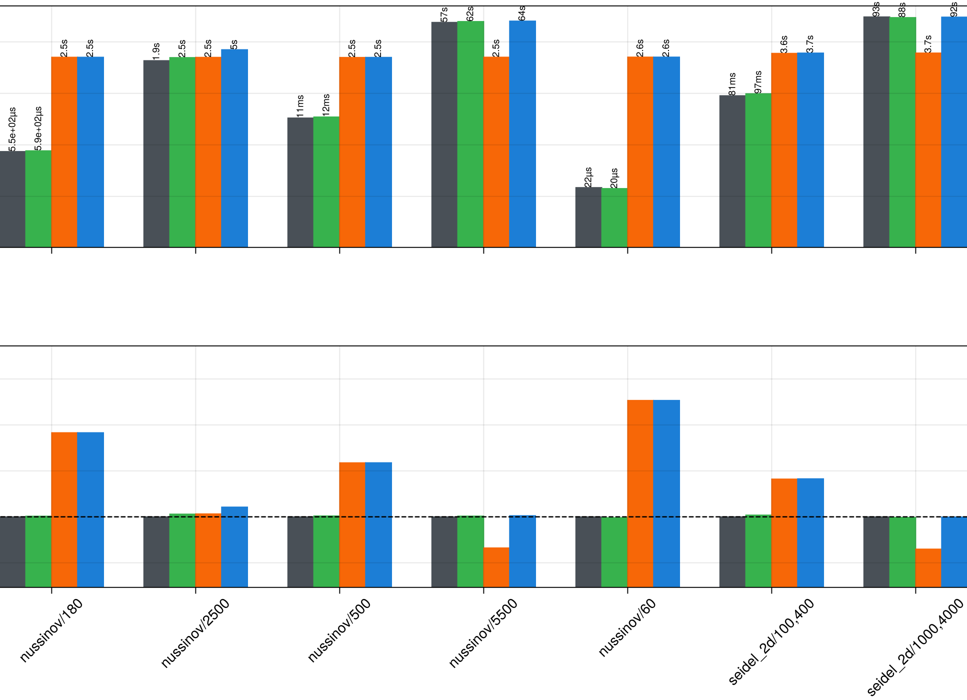

6.1ms
9.1ms
1.7s
1.7s
5.1e+02ns
5.3e+02ns
1s
1s
5.4µs
7.5µs
1s
1s
3.7s
4s
1.7s
5.6s
5.9µs
7.5µs
1.6s
1.6s
8ms
11ms
1.7s
1.7s
93s
90s
2s
92s
2e+02µs
2.7e+02µs
1.7s
1.7s
gesummv/2800
gesummv/30
gesummv/90
gramschmidt/1000,1200
gramschmidt/20,30
gramschmidt/200,240
gramschmidt/2000,2600
gramschmidt/60,80

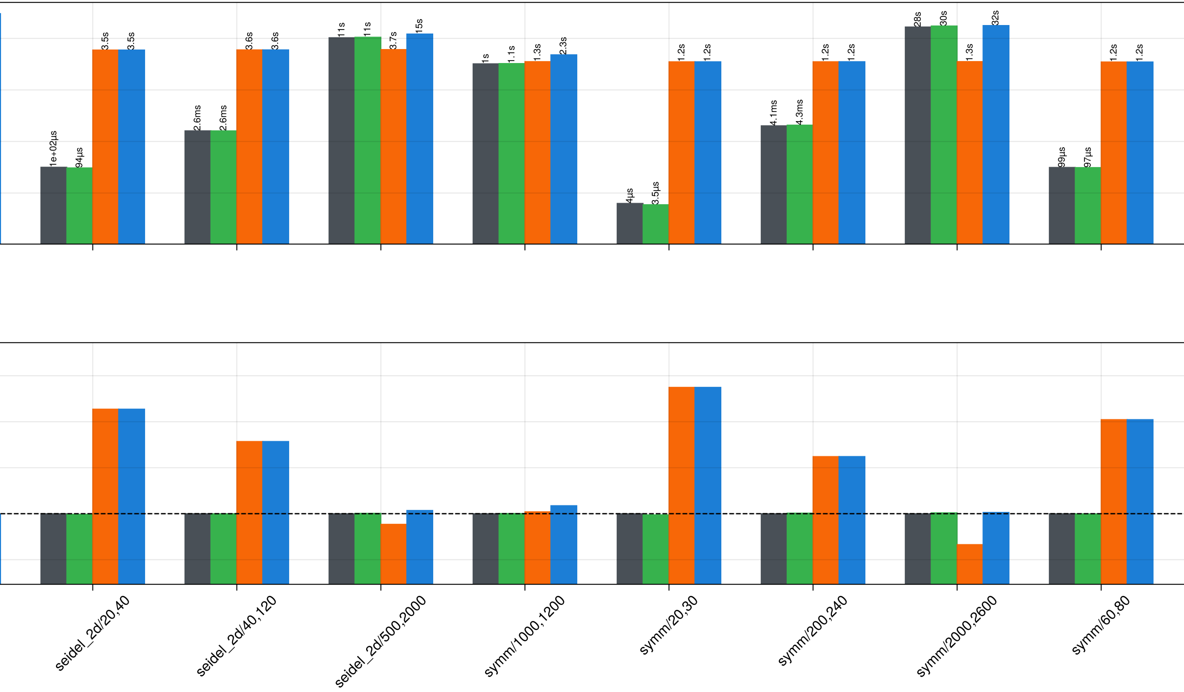

1.2e+02ms
1.2e+02ms
3.1s
3.2s
14s
13s
3s
16s
6.6e+02µs
5.3e+02µs
3s
3s
7.4ms
7.1ms
3s
3s
3.7s
4.3s
3s
7.3s
14µs
16µs
3.1s
3.1s
1.6ms
1.6ms
3.1s
3.1s
3.4e+02ns
3.2e+02ns
3.1s
3.1s
heat_3d/100,40
heat_3d/1000,200
heat_3d/20,10
heat_3d/40,20
heat_3d/500,120
jacobi_1d/100,400
jacobi_1d/1000,4000
jacobi_1d/20,30

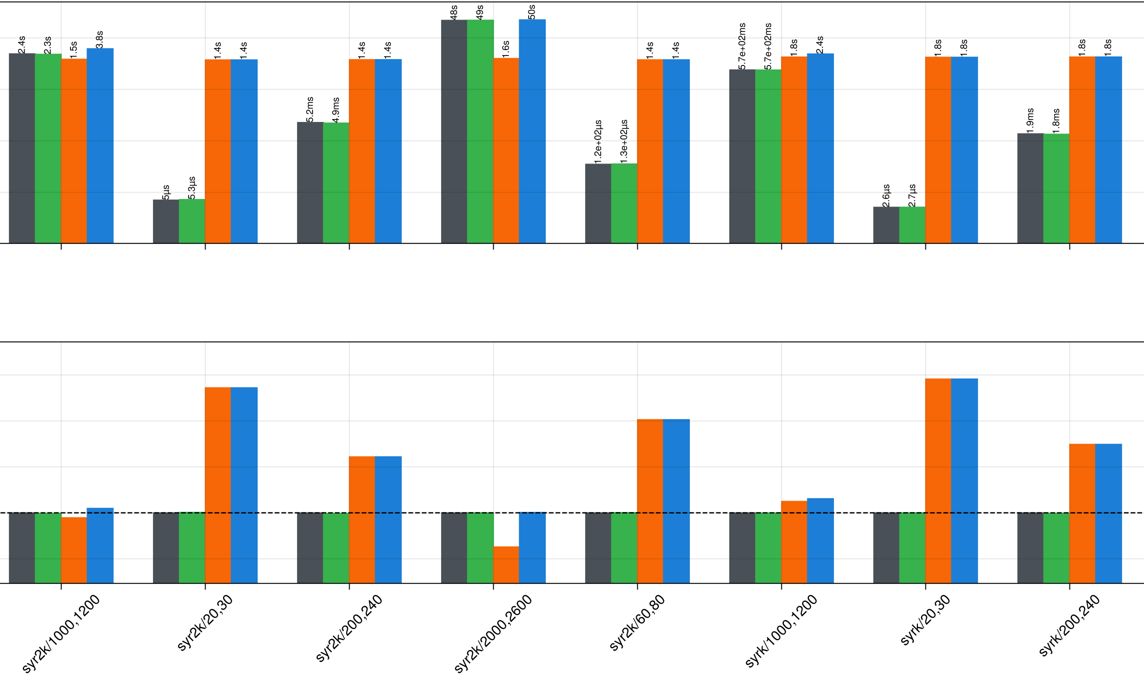

1.7µs
1.9µs
3.1s
3.1s
3.5e+02µs
3.5e+02µs
3.1s
3.1s
4.4ms
5.1ms
3.4s
3.4s
11s
11s
3.7s
15s
10µs
9.3µs
3.4s
3.4s
2e+02µs
1.9e+02µs
3.4s
3.4s
9.9e+02ms
1e+03ms
3.6s
4.6s
2.5e+02µs
jacobi_1d/40,120
jacobi_1d/500,2000
jacobi_2d/100,250
jacobi_2d/1000,2800
jacobi_2d/20,30
jacobi_2d/40,90
jacobi_2d/500,1300

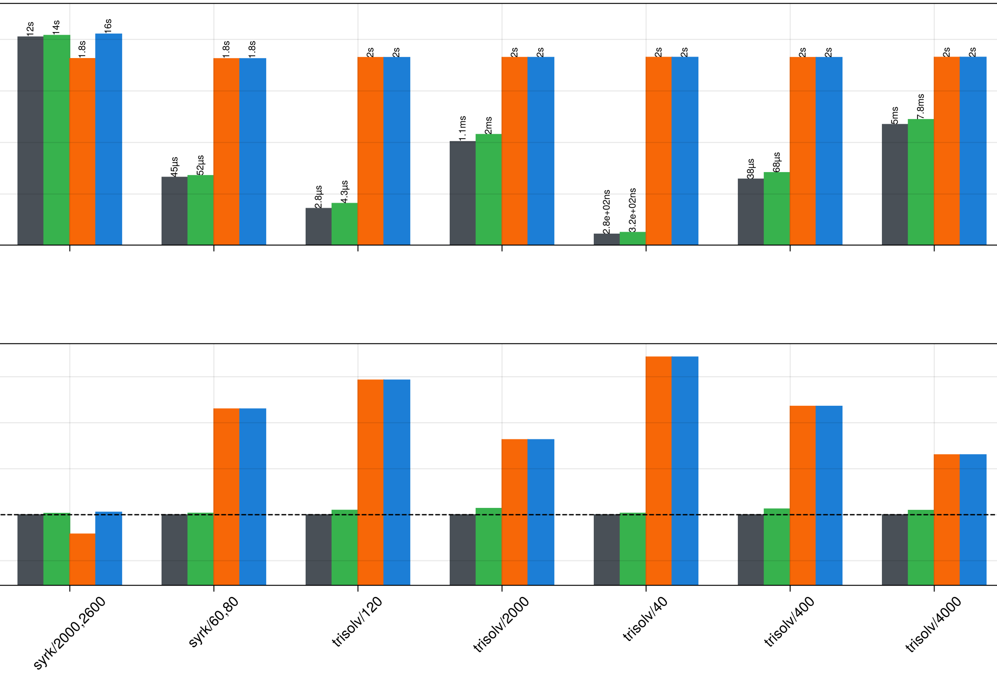

2.9e+02µs
1.9s
1.9s
2.8s
2.8s
1.9s
4.7s
9.1µs
9.2µs
1.8s
1.8s
13ms
17ms
1.9s
1.9s
60s
58s
1.9s
59s
2.2e+02µs
2.5e+02µs
2s
2s
2.5s
2.8s
2s
4.8s
8.9µs
8.2µs
2.1s
2.1s
lu/120
lu/2000
lu/40
lu/400
lu/4000
ludcmp/120
ludcmp/2000
ludcmp/40

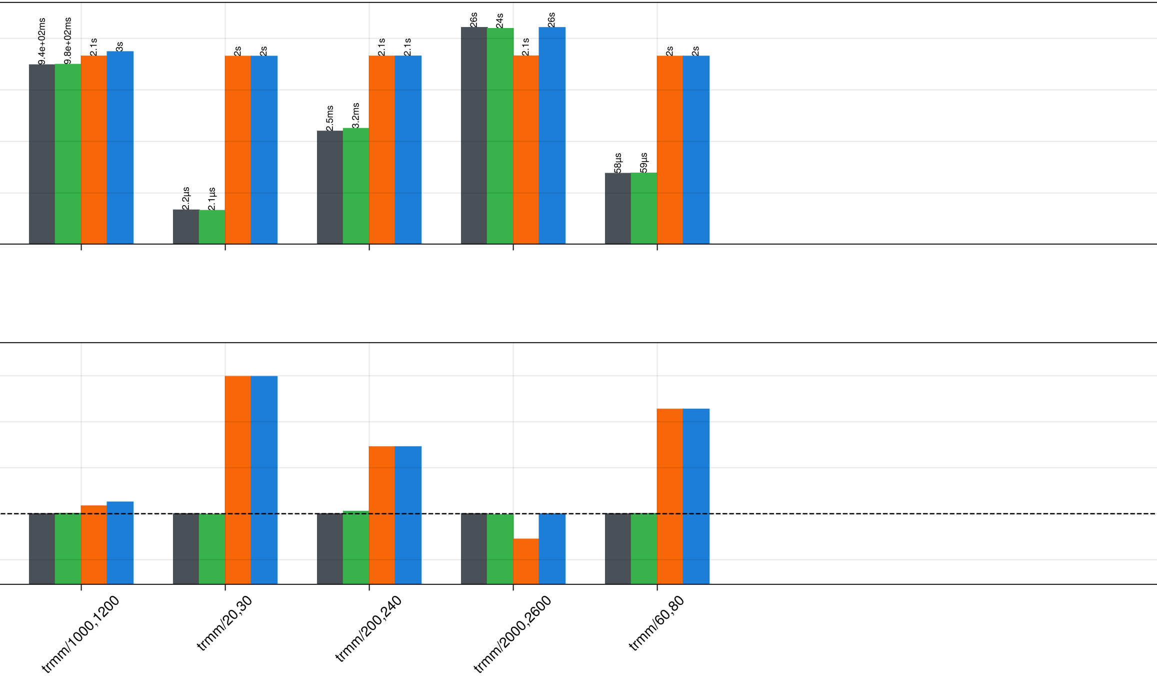

12ms
13ms
2.1s
2.1s
58s
64s
2s
66s
12µs
21µs
6.6e+02ms
6.6e+02ms
10ms
13ms
7.9e+02ms
8.1e+02ms
9.7e+02ns
1.2µs
6.6e+02ms
6.6e+02ms
2e+02µs
3.2e+02µs
6.7e+02ms
6.7e+02ms
90ms
1.1e+02ms
6.8e+02ms
7.9e+02ms
ludcmp/400
ludcmp/4000
mvt/120
mvt/2000
mvt/40
mvt/400
mvt/4000


5.5e+02µs
5.9e+02µs
2.5s
2.5s
1.9s
2.5s
2.5s
5s
11ms
12ms
2.5s
2.5s
57s
62s
2.5s
64s
22µs
20µs
2.6s
2.6s
81ms
97ms
3.6s
3.7s
93s
88s
3.7s
92s
nussinov/180
nussinov/2500
nussinov/500
nussinov/5500
nussinov/60
seidel_2d/100,400
seidel_2d/1000,4000


1e+02µs
94µs
3.5s
3.5s
2.6ms
2.6ms
3.6s
3.6s
11s
11s
3.7s
15s
1s
1.1s
1.3s
2.3s
4µs
3.5µs
1.2s
1.2s
4.1ms
4.3ms
1.2s
1.2s
28s
30s
1.3s
32s
99µs
97µs
1.2s
1.2s
seidel_2d/20,40
seidel_2d/40,120
seidel_2d/500,2000
symm/1000,1200
symm/20,30
symm/200,240
symm/2000,2600
symm/60,80


2.4s
2.3s
1.5s
3.8s
5µs
5.3µs
1.4s
1.4s
5.2ms
4.9ms
1.4s
1.4s
48s
49s
1.6s
50s
1.2e+02µs
1.3e+02µs
1.4s
1.4s
5.7e+02ms
5.7e+02ms
1.8s
2.4s
2.6µs
2.7µs
1.8s
1.8s
1.9ms
1.8ms
1.8s
1.8s
syr2k/1000,1200
syr2k/20,30
syr2k/200,240
syr2k/2000,2600
syr2k/60,80
syrk/1000,1200
syrk/20,30
syrk/200,240


12s
14s
1.8s
16s
45µs
52µs
1.8s
1.8s
2.8µs
4.3µs
2s
2s
1.1ms
2ms
2s
2s
2.8e+02ns
3.2e+02ns
2s
2s
38µs
68µs
2s
2s
5ms
7.8ms
2s
2s
syrk/2000,2600
syrk/60,80
trisolv/120
trisolv/2000
trisolv/40
trisolv/400
trisolv/4000


9.4e+02ms
9.8e+02ms
2.1s
3s
2.2μs
2.1μs
2s
2s
2.5ms
3.2ms
2.1s
2.1s
26s
24s
2.1s
26s
58μs
59μs
2s
2s
trmm/1000,1200
trmm/20,30
trmm/200,240
trmm/2000,2600
trmm/60,80